

Multi-Turn Speech Dialogue: 2018-2022 Supplement

Papers retained from readme_pre_2023.docx after removing entries published before 2018.

Scope: multi-turn spoken interaction, spoken dialogue corpora and evaluation, turn-taking/interruption, incremental or barge-in-capable systems, and training methods for spoken dialogue. Text-only dialogue papers and single-turn ASR/TTS/SLU work are excluded unless they directly model turn-taking for spoken systems.

Cutoff	Retained papers	Source filter
2018-2022	32	60 pre-2018 entries removed

Survey

1. **"Turn-taking in Conversational Systems and Human-Robot Interaction: A Review"**. Gabriel Skantze. Computer Speech & Language 2021. [\[Paper\]](#)

General Surveys

1. **"A Survey of Available Corpora For Building Data-Driven Dialogue Systems: The Journal Version"**. Iulian V. Serban et al.. Dialogue & Discourse 2018. [\[Paper\]](#)
2. **"Neural Approaches to Conversational AI"**. Jianfeng Gao et al.. Foundations and Trends in Information Retrieval 2019. [\[Paper\]](#)
3. **"Recent Neural Methods on Dialogue State Tracking for Task-Oriented Dialogue Systems: A Survey"**. Vevake Balaraman et al.. SIGDIAL 2021. [\[Paper\]](#)
4. **"Who Says What to Whom: A Survey of Multi-Party Conversations"**. Jia-Chen Gu et al.. IJCAI 2022. [\[Paper\]](#)

Datasets

1. **"Japanese Dialogue Corpus of Information Navigation and Attentive Listening Annotated with Extended ISO-24617-2 Dialogue Act Tags"**. Koichiro Yoshino et al.. LREC 2018. [\[Paper\]](#)
2. **"Taskmaster-1: Toward a Realistic and Diverse Dialog Dataset"**. Bill Byrne et al.. EMNLP-IJCNLP 2019. [\[Paper\]](#)
3. **"Alexa in the Wild: Collecting Unconstrained Conversations with a Modern Voice Assistant in a Public Environment"**. Benjamin Benk et al.. LREC 2020. [\[Paper\]](#)
4. **"HarperValleyBank: A Domain-Specific Spoken Dialog Corpus"**. Mike Wu et al.. arXiv 2020. [\[Paper\]](#)
5. **"The MSP-Conversation Corpus"**. Luz Martinez-Lucas et al.. Interspeech 2020. [\[Paper\]](#)
6. **"ASCEND: A Spontaneous Chinese-English Dataset for Code-switching in Multi-turn Conversation"**. Holy Lovenia et al.. arXiv 2021. [\[Paper\]](#)
7. **"Collection and Analysis of Travel Agency Task Dialogues with Age-Diverse Speakers"**. Michimasa Inaba et al.. LREC 2022. [\[Paper\]](#)
8. **"Design and Evaluation of the Corpus of Everyday Japanese Conversation"**. Hanae Koiso et al.. LREC 2022. [\[Paper\]](#)

Evaluation & Benchmarks

Evaluation Focus: Turn-taking & Interruption

1. **"Investigating Speech Features for Continuous Turn-Taking Prediction Using LSTMs"**. *Matthew Roddy et al.*. Interspeech 2018. [\[Paper\]](#)
2. **"Turn-Taking Predictions Across Languages and Genres Using an LSTM Recurrent Neural Network"**. *Nigel G. Ward et al.*. SLT 2018. [\[Paper\]](#)

Evaluation Focus: Full-duplex Interaction

No paper from 2018 through 2022 is listed in this subsection.

Evaluation Focus: Multi-turn Dialogue Capabilities

No paper from 2018 through 2022 is listed in this subsection.

Models

Interaction Mode: Turn-taking

1. **"Improving End-of-Turn Detection in Spoken Dialogues by Detecting Speaker Intentions as a Secondary Task"**. *Zakaria Aldeneh et al.*. ICASSP 2018. [\[Paper\]](#)
2. **"Neural Dialogue Context Online End-of-Turn Detection"**. *Ryo Masumura et al.*. SIGDIAL 2018. [\[Paper\]](#)
3. **"Prediction of Turn-Taking Using Multitask Learning with Prediction of Backchannels and Fillers"**. *Kohei Hara et al.*. Interspeech 2018. [\[Paper\]](#)
4. **"Turn-Taking Prediction Based on Detection of Transition Relevance Place"**. *Kohei Hara et al.*. Interspeech 2019. [\[Paper\]](#)
5. **"TurnGPT: A Transformer-Based Language Model for Predicting Turn-Taking in Spoken Dialog"**. *Erik Ekstedt and Gabriel Skantze*. EMNLP Findings 2020. [\[Paper\]](#)
6. **"Projection of Turn Completion in Incremental Spoken Dialogue Systems"**. *Erik Ekstedt and Gabriel Skantze*. SIGDIAL 2021. [\[Paper\]](#)
7. **"Timing Generating Networks: Neural Network Based Precise Turn-Taking Timing Prediction in Multiparty Conversation"**. *Shinya Fujie et al.*. Interspeech 2021. [\[Paper\]](#)
8. **"Gated Multimodal Fusion with Contrastive Learning for Turn-Taking Prediction in Human-Robot Dialogue"**. *Jiudong Yang et al.*. ICASSP 2022. [\[Paper\]](#)
9. **"Response Timing Estimation for Spoken Dialog System Using Dialog Act Estimation"**. *Jin Sakuma et al.*. Interspeech 2022. [\[Paper\]](#)
10. **"Turn-Taking Prediction for Natural Conversational Speech"**. *Shuo-Yiin Chang et al.*. Interspeech 2022. [\[Paper\]](#)
11. **"Voice Activity Projection: Self-Supervised Learning of Turn-Taking Events"**. *Erik Ekstedt and Gabriel Skantze*. Interspeech 2022. [\[Paper\]](#)

Interaction Mode: Half-duplex / Controlled Barge-in

1. **"An Incremental Turn-Taking Model for Task-Oriented Dialog Systems"**. *Andrei C. Coman et al.*. Interspeech 2019. [\[Paper\]](#)
2. **"Oh, Jeez! or Uh-Huh? A Listener-Aware Backchannel Predictor on ASR Transcriptions"**. *Daniel Ortega et al.*. ICASSP 2020. [\[Paper\]](#)
3. **"BPM_MT: Enhanced Backchannel Prediction Model Using Multi-Task Learning"**. *Jin Yea Jang et al.*. EMNLP 2021. [\[Paper\]](#)

4. **"Contextual Acoustic Barge-In Classification for Spoken Dialog Systems"**. *Dhanush Bekal et al.*. Interspeech 2022. [\[Paper\]](#)

Interaction Mode: Full-duplex

No paper from 2018 through 2022 is listed in this subsection.

Training Methods

Interaction Mode: Turn-taking

1. **"Towards Generalized Models for Task-Oriented Dialogue Modeling on Spoken Conversations"**. *Ruijie Yan et al.*. arXiv 2022. [\[Paper\]](#)

Interaction Mode: Half-duplex / Streaming

1. **"A Methodology for Turn-Taking Capabilities Enhancement in Spoken Dialogue Systems Using Reinforcement Learning"**. *Hatim Khouzaimi et al.*. Computer Speech & Language 2018. [\[Paper\]](#)

Interaction Mode: Full-duplex

No paper from 2018 through 2022 is listed in this subsection.